Sardar Patel Institute of Technology

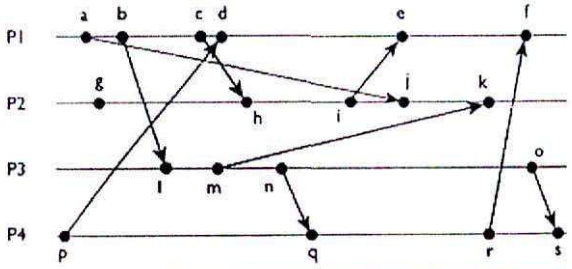
Bhavan's Campus, Munshi Nagar, Andheri (W), Mumbai : 400058, India

(Empowered Autonomous College of Affiliated to University of Mumbai)

End Semester Examination

December 2023

Max Marks: 100 Duration: 3 hours
Class: T.Y. B.Tech Semester: V
Course code: CS304 Branch: Computer Engg.
Name of the course: Distributed Computing

Q No		Max Marks	CO	BL
Q.1 (a)	Discuss relative advantages and various commonly used models for configuring different distributed systems. Which model do you think is going to become the most popular model in future – give reason for your answer?	10	1	2
Q.1 (b)	Give the Vector Clock timestamps for the following event diagram 	10	4	4
Q. 2(a)	What were the major economic, social factors that motivated the development of distributed computing systems? What are some of the main advantages and disadvantages of distributed computing systems over centralized systems?	10	1	3
Q 2(b)	Illustrate the architecture of a D'agent and elaborate the implementation issues.	10	2	4
Q. 3(a)	What are distributed objects, show how organization occurs of remote object with client-side proxy? Compare Compile-time v/s run-time objects and give drawbacks of each.	10	2	2

Q 3 (b)	What purpose does Election Algorithms serve? Write the working of Ring Algorithm with an example. **OR** With the help of neat system diagrams, describe the following load balancing Algorithms – 1. Round Robin 2. Least Connections	10	3	2
Q.4(a)	Describe Ricart-Agrawala's Algorithm for distributed mutual exclusion, include relevant diagrams and highlight its advantages and disadvantages over other Non-Token based algorithms.	10	3	3
Q.4(b)	With the help of timing diagram, describe the following consistency models – 1. Causal Consistency 2. Release Consistency	10	3	4
Q.5(a)	What are different data replication schemes, give advantages and disadvantages of the same with proper system diagrams.	10	3	4
Q.5(b)	With the help of system diagrams, describe various consistency protocols.	10	3	3